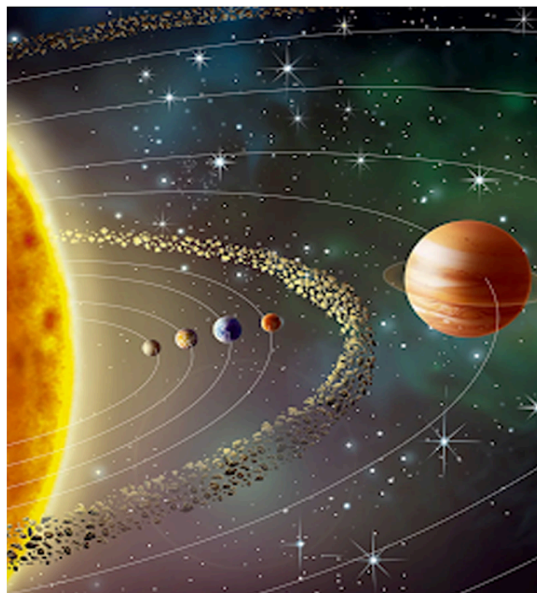
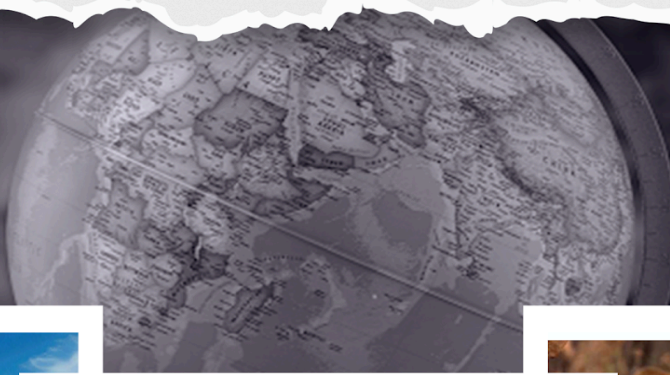


General Knowledge

World Geography

STUDY NOTES

Composition and Structure of Atmosphere



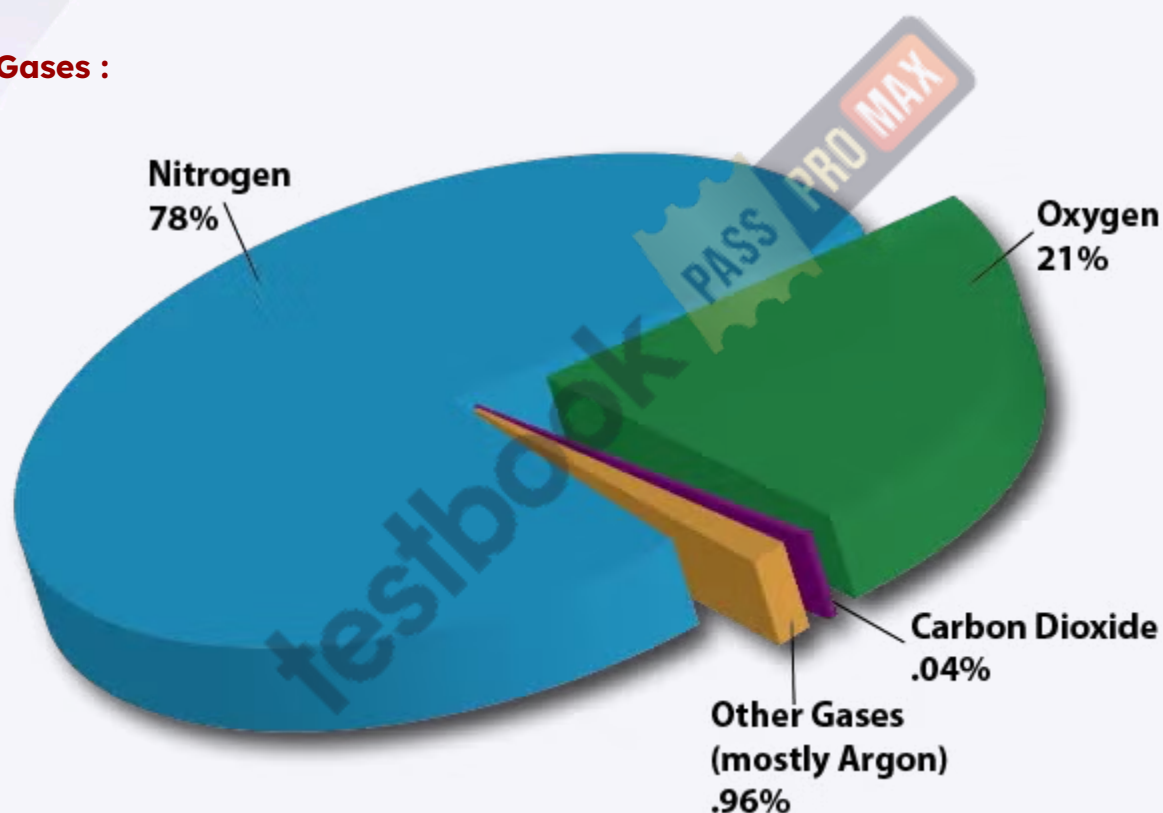
Climatology: Composition and Structure of Atmosphere

The atmosphere is a thick **gaseous envelope** that surrounds the earth from all sides and is attached to the Earth's surface by gravitational force.

It provides the necessary gases for sustenance of life and **filters incoming solar radiation**. Without the atmosphere, Earth would be barren like the moon. The height of the atmosphere is around **16 to 29 thousand kilometers from the sea level**.

The atmosphere is a mechanical mixture of:

Gases :



GAS	COMPOSITION (in percentage)
Nitrogen	78

Oxygen	21
Argon	0.93
Carbon dioxide	0.03
Neon	0.0018
Helium	0.0005
Ozone	0.00006
Hydrogen	0.00005
Krypton	Trace
Xenon	Trace
Methane	Trace

- **Oxygen** is the most important of all gases from the standpoint of **survival of living organisms** essential for their breathing.
- It is also a must for the **combustion** of burning matter.
- **Nitrogen** acts as a **diluent** and is generally chemically inactive.
- **Carbon dioxide** is used by green plants for **photosynthesis** and also absorbs most radiant energy from the Earth and re-radiates it back to the Earth.

- That is why it is called a **greenhouse gas**. It increases the temperature of the lower atmosphere and the earth's surface.
- **Ozone gas** absorbs most of the **UV rays** from the sun and prevents the Earth from becoming too hot.

Water Vapour:

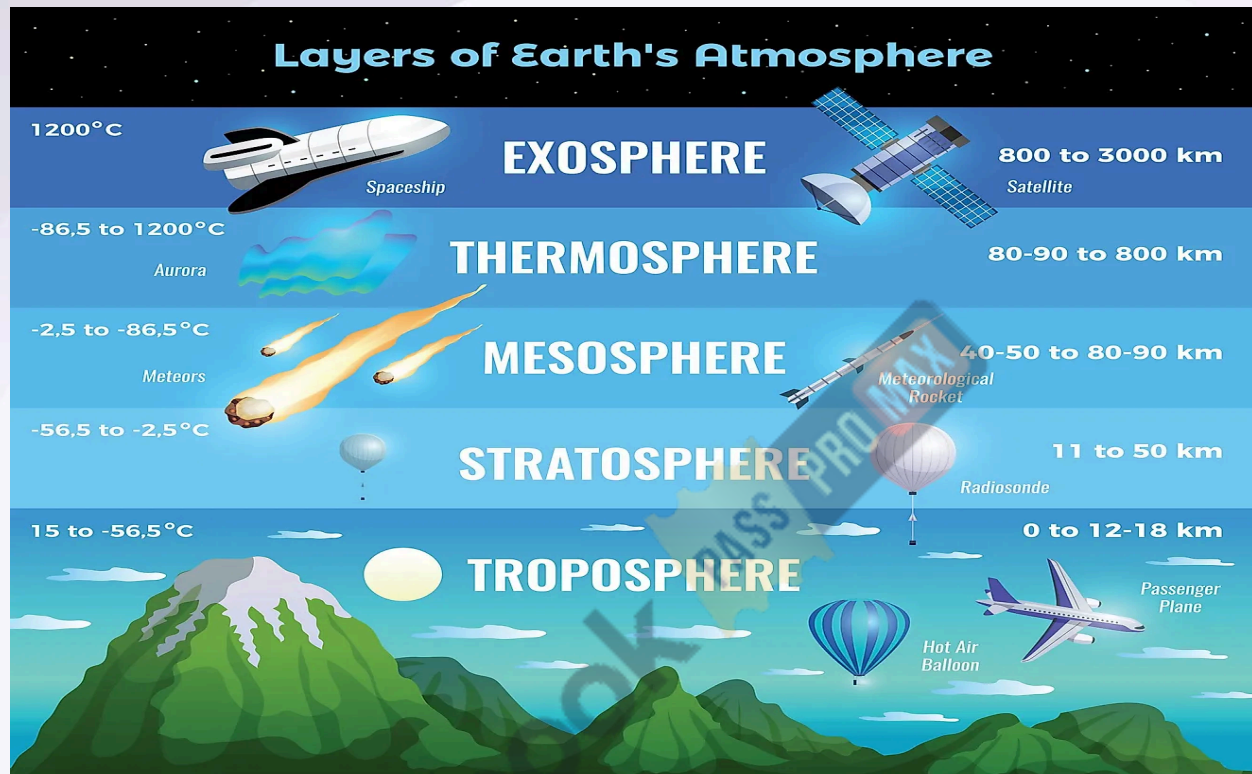
- Water vapor, an important component climatically, ranges from **0 to 5% by volume** in the atmosphere.
- It is received through the **evaporation of moisture** from water bodies, vegetation, and soil cover.
- Vapour depends on **temperature** and hence it **decreases from the equator poleward** in response to decreasing temperature towards the poles.
- The moisture content in the atmosphere creates **several forms of condensation** like clouds, fog, dew, rainfall, frost, hailstorm, snow, etc.
- Water vapor is almost **transparent for the incoming short-wave solar radiation** but is **less transparent for outgoing long-wave radiation**.
- Thus it helps in **heating the earth's surface** and the lower atmosphere because it absorbs terrestrial radiation.

Particulate Matter:

- This includes dust particles, salt particles, pollen, smoke and soot, volcanic ashes, etc. which are kept in suspension in the atmosphere.
- They help in **scattering of solar radiation** which adds varied colours of red and orange at sunrise and sunset.
- The **blue colour of the sky** is due to the **selective scattering** of solar radiation by dust particles.
- **Salt particles** become **hygroscopic nuclei** and help in the formation of **water drops**, clouds, and other forms of precipitation and condensation.

Structure of the Atmosphere:

Based on information received from rockets, radars, and satellites, the Earth's atmosphere consists of 4 zones or layers like spherical shells:



Troposphere:

- The **lowermost layer** of the atmosphere with an average height of 13 km (16 km over the equator and 6 km over the poles).
- The air we breathe exists here. Almost all **weather phenomena** (like rainfall, fog, clouds, dew, frost, and hailstorms) occur in this layer.
- Temperature decreases with increasing height at the rate of 6.5°C per 1000m. This is called the **normal lapse rate**.
- The height of the troposphere varies from the equator towards the pole (decreases towards the poles) and from one season to another (increases during summer while decreases during winter) during a year.

- The upper limit is called the **tropopause** which is 1.5 km thick. (The word “troposphere” literally means “zone/region of mixing” and tropopause means “where the mixing stops”.)

Stratosphere:

- Above the troposphere lies the stratosphere. It extends **up to a height of 50 km**.
- This layer is almost free from clouds and associated weather phenomena, making conditions **most ideal for flying aeroplanes**.
- One important feature of the stratosphere is that it contains a **layer of ozone gas** between the **height of 15 to 30 km** (also called **Ozonosphere**).
- Temperature increases with height due to the absorption of UV rays by the ozone layer.
- The upper limit of the stratosphere is known as **stratopause**.

Mesosphere:

- Lying above the stratosphere this is the third layer of the atmosphere and extends between **50 km and 80 km**.
- **Meteorites burn up** in this layer on entering from space.
- Temperature decreases with increasing height again.
- The uppermost limit is the **mesopause** where the temperature becomes -80°C .

Thermosphere:

- The temperature here rises very rapidly with increasing height.
- The **International Space Station** is situated in this layer.
- It is further divided into two layers:

Ionosphere:

- It extends between **80 - 400 km**.
- This layer helps in **radio transmission**. In fact, radio waves transmitted from the earth are reflected back to the earth by this layer.

- Includes a number of **ionic layers** with increasing heights like the D layer, E layer, F layer, and G layer.

Exosphere:

- This is the **uppermost layer** of the atmosphere.
- This layer has very thin air.
- Light gases like helium and hydrogen float into space from here.
- The **density here is very low** and the atmosphere resembles a **nebula** because it is highly rarefied.

Altitude vs Temperature

